

Chain $\leftrightarrow$ positive-root cross-check — does $\{\text{rank}, N_c, n_C, g\}$ correspond to the 4 positive roots of B_2 ? COUNT yes (forced), VALUE-bijection NO. Honest negative that pins the cyclotomic route's true strength.

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Chain $\leftrightarrow$ positive-root cross-check

0. The ask (Keeper)

Keeper: keep the cyclotomic generation route alive; if the tube number pins $\neq 3$, it carries the generation count via $\text{generations} = |\Phi^+(B_2)| - 1 = 3$. He asked me to nail the chain $\leftrightarrow$ positive-root correspondence: is the substrate commitment chain $\{\text{rank}, N_c, n_C, g\} = \{2, 3, 5, 7\}$ in structural bijection with the 4 positive roots of B_2 ? If yes, $\text{generations} = |\Phi^+| - 1$ rests on a forced structure, not a count-coincidence.

1. The test (computed)

- B_2 positive roots (convex order from $w_0 = s_1 s_2 s_1 s_2$): α_1 (ht 1, short), $2\alpha_1 + \alpha_2$ (ht 3, long), $\alpha_1 + \alpha_2$ (ht 2, short), α_2 (ht 1, long). $|\Phi^+(B_2)| = 4$ — a FORCED fixed root-system fact.
- Chain values $\{2, 3, 5, 7\}$ = the first four primes (each an independently-derived substrate primary).

level	COUNT	VALUE
chain	4	$\{2, 3, 5, 7\}$ (primes)
$\Phi^+(B_2)$	4	heights $\{1, 1, 2, 3\}$; lengths $\{\text{short}, \text{long}, \text{short}, \text{long}\}$

2. Result — honest negative

- COUNT: MATCH, and forced. $|\text{chain}| = 4 = |\Phi^+(B_2)|$, and $|\Phi^+(B_2)| = 4$ is a fixed fact (no theorem-scope issue, unlike the tube number). This is the robust ingredient.
- VALUE-bijection: FAILS. Root heights $\{1, 1, 2, 3\} \neq \{2, 3, 5, 7\}$; the length split (2 short, 2 long) is not 4 distinct primes. No natural root invariant reproduces the chain values. The chain values are primes (each derived as a substrate primary), NOT functions of the root vectors.
- Aside (flagged as coincidence-risk, NOT claimed): B_2 's exponents are $\{1, 3\}$, degrees $\{2, 4\}$, $h=4$, $h^\vee=3$ — so $N_c=3$ appears as an exponent and $\text{rank}=2$ as a degree. Suggestive, but it does NOT assemble into a clean chain $\leftrightarrow$ root map, and pattern-matching here is exactly where Cal #27 fires. I flag it and do not claim it.

3. What this means for the cyclotomic route

The cyclotomic generation route — $\text{generations} = |\Phi^+(B_2)| - 1 = 3$ — rests on exactly two ingredients: 1. $|\Phi^+(B_2)| = 4$ — FORCED (fixed root-system fact). 2. base/seed = rank (Gate 1, the $X=\text{rank}$ argument).

It does NOT rest on a chain $\leftrightarrow$ positive-root value-bijection — because there isn't one. The “chain has 4 levels” matches “ $|\Phi^+|=4$ ” as a COUNT, and a candidate structural reason exists (the 4 commitments $\leftrightarrow$ the 4 generators of the maximal nilpotent n^+ / the Hall algebra's root-vector degrees of freedom) — but that reason rides on the same $A_{\text{sub}} \leftrightarrow H$ dictionary bet (Phase 2) as everything else. So:

The cyclotomic route's forced strength = the forced COUNT $|\Phi^+|=4$ (cleaner than the tube number, which is NOT forced). It does NOT gain extra independent force from a chain $\leftrightarrow$ root structure. The “why exactly $|\Phi^+|$ commitments” link is the Phase-2 dictionary bet.

This is a useful NEGATIVE: it prevents dressing up the cyclotomic route as stronger than it is. Generation-forcing stays OPEN; the cyclotomic $|\Phi^+|-1$ route is the better-footed CARRIER (forced count) but is not itself a closed forcing.

4. Honest scope + tier

RIGOROUS: $|\Phi^+(B_2)| = 4$ (forced); the value-bijection failure (computed).

NEGATIVE RESULT (this doc): no chain $\leftrightarrow$ positive-root value-bijection; the cyclotomic route rests on $|\Phi^+|=4$ (forced) + base/seed (Gate 1), not on a chain $\leftrightarrow$ root structure.

OPEN: generation-forcing (the count = 3 is matched/over-determined; the cyclotomic carrier is forced at the COUNT but the “why $|\Phi^+|$ commitments” mechanism rides on the Phase-2 dictionary). Neither tube nor cyclotomic route is a closed forcing.

Cal #27 / honesty: this is a case of the discipline producing a clean negative — I went looking for a chain $\leftrightarrow$ root structure to strengthen a favored route, did not find one at the value level, and am reporting that rather than pattern-matching the exponent/degree coincidence into a claim. The route is exactly as strong as $|\Phi^+|=4$ + Gate 1, no stronger.

Next (my lane): the Phase-2 $A_{\text{sub}} \leftrightarrow H(Q_{B2})$ dictionary — which is also where the “commitments $\leftrightarrow n^+$ generators” structural reason would be established (the same gate). The dictionary is the common bottleneck for the physical identifications AND the generation-count mechanism.

— Lyra, chain $\leftrightarrow$ positive-root cross-check v0.1. HONEST NEGATIVE: the correspondence holds at the COUNT level ($4 = |\Phi^+(B_2)|$, FORCED) but FAILS at the VALUE level (chain $\{2,3,5,7\}$ are primes; root heights $\{1,1,2,3\}$ — no bijection; exponent/degree near-matches flagged as coincidence-risk, not claimed). So the cyclotomic generation route rests on $|\Phi^+|=4$ (forced) + base/seed (Gate 1), NOT on a deeper chain $\leftrightarrow$ root structure; the “why $|\Phi^+|$ commitments” link rides on the Phase-2 dictionary. Prevents an overclaim. Generation-forcing stays OPEN; cyclotomic is the better-footed CARRIER (forced count) but not a closed forcing.